Expense Tracker & Budget Manager

A command-line personal finance application built in Python, developed as the evaluated project for the **Python Essentials** course.

Overview

Expense Tracker & Budget Manager is a lightweight, dependency-free CLI tool for logging personal expenses, setting category-wise monthly budgets, and generating spending reports — all backed by a local JSON file, with no database server or GUI required. The project applies core course concepts including functions, object-oriented design, exception handling, file I/O, and modular program structure.

Features

- **Expense Management** — add, list (filter by category and month), update, and delete expense records
- **Budget Management** — set/update a monthly spending limit per category, with automatic warnings when a budget is exceeded
- **Reporting & Analytics**
  - Category-wise spending breakdown (text-based bar chart)
  - Month-by-month summary
  - Top-5 largest expenses
- **Input Validation** — amount, category, description, and date are validated before being saved
- **Atomic File Writes** — writes to a temp file and uses `os.replace()` to prevent data corruption if the program is interrupted
- **Custom Exception Hierarchy** — `ValidationError`, `ExpenseNotFoundError`, `BudgetExceededError`, `StorageError`
- **Logging** — all state-changing actions and errors are timestamped and written to `app.log`
- **Simple Menu-Driven Interface** — numbered menu, clear prompts, plain-language error messages (no raw stack traces)

Technologies / Tools Used

- **Python 3** (standard library only — no external dependencies)
- `json` — data persistence
- `dataclasses` — `Expense` and `Budget` models
- `unittest` — automated testing
- `logging` — action/error logging
- **Git** — version control

Project Structure

```
expense_tracker/
├── cli.py          # Menu-driven CLI (only module that calls input()/print())
├── manager.py      # ExpenseManager — add/list/update/delete expenses
├── budget.py       # BudgetManager — set/check category budgets
├── reports.py      # ReportGenerator — category, monthly, top-5 reports
├── storage.py      # JSONStorage — centralizes all file I/O
├── models.py       # Expense and Budget dataclasses
├── utils.py        # Shared input validation logic
├── exceptions.py   # Custom exception hierarchy
├── data/
│   └── expenses.json
├── app.log
tests/
├── test_manager.py
```

```
├── test_budget.py
└── test_reports.py
main.py             # Entry point
```

Steps to Install & Run

1. **Clone the repository**

   ```
   git clone <repository-url>
   cd expense_tracker
   ```

2. **Requirements**: Python 3.8+ (no external packages needed)

3. **Run the application**

   ```
   python3 main.py
   ```

4. **Use the menu** to add expenses, set budgets, and view reports:

   ```
   ==================== EXPENSE TRACKER ====================
   1. Add expense
   2. List expenses
   3. Update expense
   4. Delete expense
   5. Set / update a category budget
   6. View budget status (current month)
   7. Category spending report
   8. Monthly summary report
   9. Top 5 expenses
   0. Exit
   ==========================================================
   ```

Instructions for Testing

The project uses Python's built-in `unittest` framework with 17 automated tests across three modules, each run against an isolated temporary JSON file so tests never touch real user data.

Run all tests:

```
python3 -m unittest discover -s tests -v
```

Test coverage: - `test_manager.py` (7 tests) — ID assignment, category/month filtering, not-found handling, update, delete, total calculation - `test_budget.py` (5 tests) — budget creation/update, over-budget detection, within-budget pass-through, multi-category status - `test_reports.py` (5 tests) — category breakdown totals, monthly grouping, top-expense ordering, bar-chart rendering (incl. empty-data case)

Manual end-to-end testing was also performed by piping a scripted sequence of menu choices into the CLI to confirm the full add → budget → report workflow.

Screenshots

Adding an expense:

```
Choose an option: 1
Categories: Food, Transport, Housing, Utilities, Entertainment, Health, Shopping, Education, Other
Amount: 250
Category: Food
Description: Groceries at market
Date (YYYY-MM-DD, blank = today):
Added: [ 1] 2026-09-26 Food 250.00 Groceries at market
```

Category spending report:

```
Food       | ######################################## 250.00
Transport  | ############                              80.00
```

Future Enhancements

- Export reports to CSV or PDF
- Multi-currency support with conversion
- Recurring/scheduled expenses (e.g. monthly rent)
- Optional SQLite storage backend for larger datasets

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